NBA Shot Quality Capstone

Exploratory Data Analysis

DAT 490 — Data Science Capstone (2026 Summer A)

Team Members

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Exploratory Data Analysis

Data sources and setup

The data for this project comes from the NBA Stats API (accessed through the open-source nba_api Python package) supplemented by two Kaggle datasets. The NBA Stats API gives us shot-level detail for every field goal attempt from the 2013-14 season through the current 2025-26 season — roughly two million shots in total — and includes the matching game logs, player bios, and team metadata. The two Kaggle supplements are the 2014-15 NBA Shot Logs dataset (dansbecker/nba-shot-logs, about 128,000 shots used for early prototyping) and the NBA Stats 1947-present dataset (sumitrodatta/nba-aba-baa-stats, used for long-term player history). All sources map to NBA.com identifiers so joins are ID-based rather than name-based.

After the data pull, we combined the thirteen seasons of shot data into a single table with an added SEASON column derived from the file name. The combined raw shot table covers every field goal attempt across the tracking-data era. The dependent variable we care about is SHOT_MADE_FLAG, a binary indicator of whether a given shot went in.

Variables analyzed

The variables we focused on for this exploratory pass are the ones most directly related to whether a shot is likely to go in. They are:

- SHOT_MADE_FLAG — dependent variable, binary (1 if the shot was made, 0 if missed)
- SHOT_DISTANCE — distance from the basket in feet
- LOC_X, LOC_Y — court coordinates of where the shot was taken
- SHOT_ZONE_BASIC — categorical zone label (Restricted Area, In The Paint Non-RA, Mid-Range, Left Corner 3, Right Corner 3, Above the Break 3, Backcourt)
- SHOT_TYPE — either "2PT Field Goal" or "3PT Field Goal"
- PERIOD — quarter of the game
- ACTION_TYPE — descriptive shot type (jump shot, layup, dunk, etc.)
- SEASON — derived from the file name; identifies which year a shot is from

In addition to these raw variables, we derived a few simple summary measures during the analysis itself: per-season averages for shot distance, share of attempts by zone, and field goal percentage by zone. These are not new columns added to the table, they're aggregations we computed for the visualizations that follow.

How shot selection has changed over time

The first thing worth looking at is how the league's shot selection has changed across the tracking-data era, because the answer is dramatic. Appendix A shows the average shot distance per season from 2013-14 through 2025-26. The trend is steadily upward through the late 2010s and then plateaus in the most recent seasons, consistent with a league that gradually moved further from the basket and then settled into a new normal.

Appendix B breaks the same trend out by shot type. The share of attempts that are three-pointers rises sharply over the era, while the share of two-point attempts falls by a near-equal amount. This is the "three-point revolution" that's been written about extensively in the analytics literature, and our data shows it very clearly. Appendix C then plots the field goal percentages for two-point and three-point attempts separately. Interestingly, the FG% for each shot type stays fairly flat over the era, even as the volumes shift so dramatically. The league as a whole is not getting better at three-pointers — it's just taking more of them.

Appendix D zooms in further by showing the share of attempts taken in each shot zone over time. The two most visible movements are above-the-break threes (rising sharply) and mid-range shots (falling just as sharply). Over our full tracking era, mid-range attempts dropped from about 27 percent of all attempts in 2013-14 to about 10 percent in 2025-26; a 17 percentage point drop. Above-the-break threes went the opposite direction, rising from about 19 percent to about 31 percent. Restricted area attempts dipped slightly; these shifts are the clearest evidence in the entire EDA that shot location is not a static feature of the data we're modeling, and that any model trained across the full era has to be able to handle this distributional change.

Which zones are efficient

Looking at field goal percentage by zone aggregated across the full era gives us the efficiency landscape we're modeling against. The numbers (visualized in Appendix E) are roughly:

Zone	FG% (2013–present aggregate)
Restricted Area	63.6%
In The Paint (Non-RA)	42.1%
Mid-Range	40.5%
Right Corner 3	38.7%
Left Corner 3	38.6%
Above the Break 3	35.2%
Backcourt	2.7%

The Restricted Area is overwhelmingly the most efficient place to shoot, followed by a cluster of two-point and corner-three shots that all sit in the high-30s to low-40s, followed by above-the-break threes in the mid-30s. The Backcourt zone is noise, as it's mostly heaves and buzzer-beaters and doesn't behave like a normal shot.

Appendix F plots this same FG% per zone but broken out by season, and the most interesting result is that the relative ranking of zones stays almost perfectly stable across the entire 12-year window. Within zones, though, there are real changes; most notably, Restricted Area FG% rises from about 61% in 2013-14 to about 67% in 2025-26 (a six percentage point increase), and In-the-Paint Non-RA also climbs by about five percentage points. Corner threes and above-the-break threes are flat in FG% over the era. The takeaway is that players are not getting better at the three-point shot, they are getting better at finishing at the rim — even as they take fewer shots from inside the arc.

How attempts and FG% changed from 2013-14 to 2025-26

To make the era shift concrete, we computed a direct change table comparing the first season of the tracking era to the most recent season. The result:

Zone	Attempt % 2013-14	Attempt % 2025-26	Δ	FG% 2013-14	FG% 2025-26	Δ
Above the Break 3	19.2%	30.7%	+11.5	35.3%	34.9%	-0.3
Mid-Range	26.8%	10.1%	-16.7	39.5%	41.6%	+2.1
Restricted Area	32.3%	28.4%	-3.9	60.9%	67.0%	+6.1
In The Paint (Non-RA)	14.9%	20.0%	+5.1	39.3%	44.4%	+5.0
Right Corner 3	3.1%	5.2%	+2.0	38.8%	38.4%	-0.4
Left Corner 3	3.5%	5.6%	+2.0	39.2%	38.7%	-0.5

The pattern is clear: the league moved its attempt volume from the mid-range to above-the-break and corner threes, and the underlying efficiency of those threes did not improve, while the efficiency of the shots taken at the rim improved.

Appendix G shows the same era shift visually — a side-by-side scatter plot of 10,000 sampled shot locations from 2013-14 alongside 10,000 from 2025-26. The 2026 plot has a much denser concentration of dots beyond the three-point arc and a visibly thinner mid-range band compared to 2013.

Winning teams versus losing teams

We also looked at how the shot profiles of winning teams differ from losing teams, joining each shot to its game's win/loss outcome. The aggregated 2013-2026 picture is that winning teams shoot much better in every single zone, with the gaps clustered around four to six percentage points. The largest gaps appear in the three-point zones (above-the-break threes are 5.85 points higher for winning teams; corner threes are 6.25 and 6.46 points higher), and the smallest gap is in the restricted area at 4.30 points.

A more interesting follow-up is whether that winning-team advantage has changed over time. Comparing 2013-14 to 2025-26 directly, the win/loss FG% gap has actually *narrowed* in most zones — for example, the above-the-break three gap shrank from 6.21 in 2013-14 to 4.96 in 2025-26, and the mid-range gap shrank from 3.98 to 2.65. This is consistent with the broader idea that league-wide shooting talent has converged: as more teams have built around modern shooting, the difference between winning and losing teams from any specific spot has gotten smaller, even as the value of the attempts themselves has shifted dramatically. This is an interesting finding because it suggests that the gap between good and bad teams in our era is less about shot success rate within zones and more about which zones each team chooses to take their shots from.

Implications for the model

Several findings from this EDA shape how we're approaching the modeling work.

First, shot location is the single most informative feature for shot success — which the EDA confirms and which Marc's initial logistic regression baseline already corroborated (his strongest feature by coefficient magnitude was SHOT_DISTANCE, negative; his strongest positive was CLOSE_DEFENDER_DISTANCE). Any expected-shot-value model has to capture spatial features explicitly.

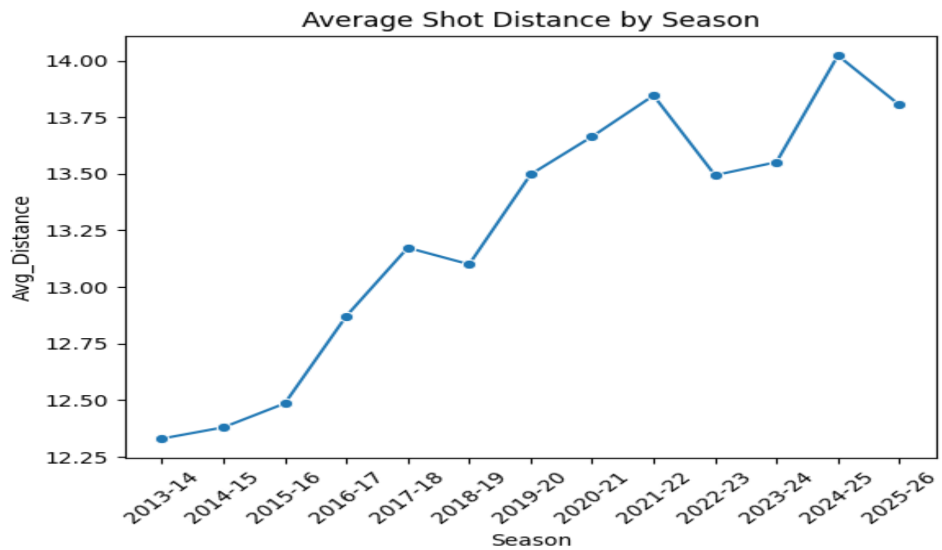
Second, the league-wide attempt distribution has shifted dramatically over the tracking era, but the within-zone efficiency rankings have stayed stable. This means a model trained on the full 2013-present window can use zone-level features without worrying that the *value* of being in a given zone has changed, even though the *frequency* of being in each zone has shifted.

Third, the winning-team advantage in shot success is smaller and more uniform than we expected, which strengthens the case for our RQ1 approach of using residual analysis to identify players and teams that exceed model expectations. If winning teams were simply better at making any given shot, residuals would be noisy; the fact that the gap is moderate and varies by zone means residual-based player evaluation can carry good signal.

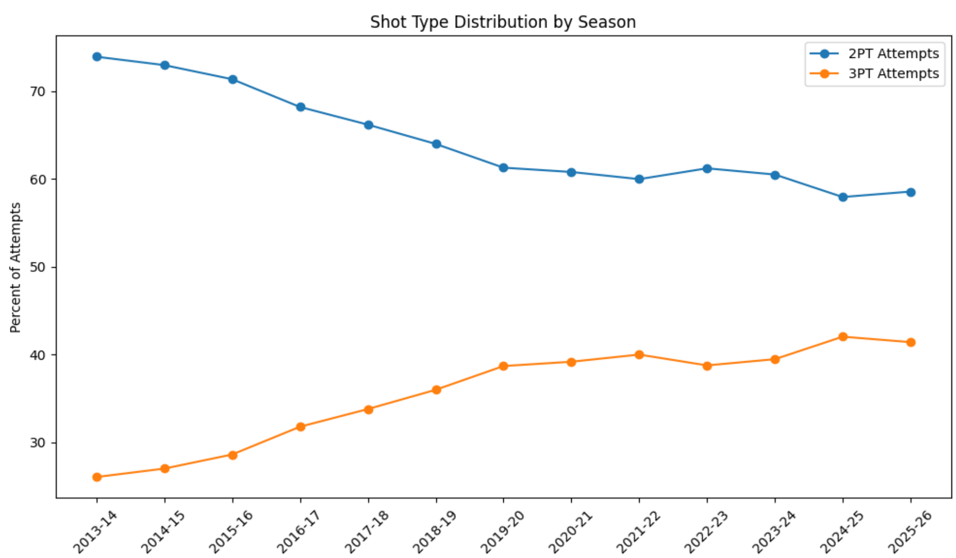
Finally, this EDA confirms RQ2 has substantive findings to surface: the geography of shot success has clearly evolved across the tracking era, and that evolution is rich enough to support good analysis rather than a flat trend.

Appendix

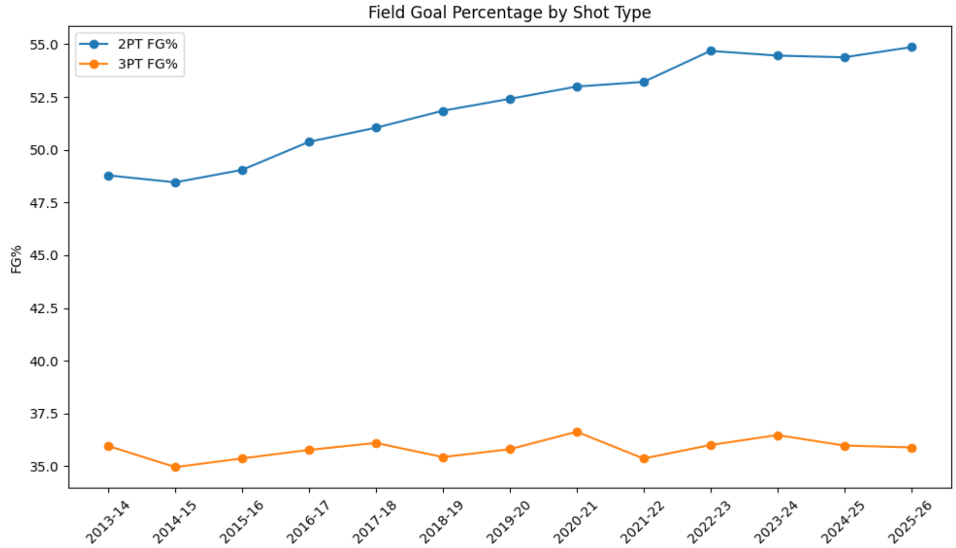
The figures below are the visualizations referenced in the prose above. Source code (including the queries and data transformations behind each chart) will be included in the Appendix of the Final Report.



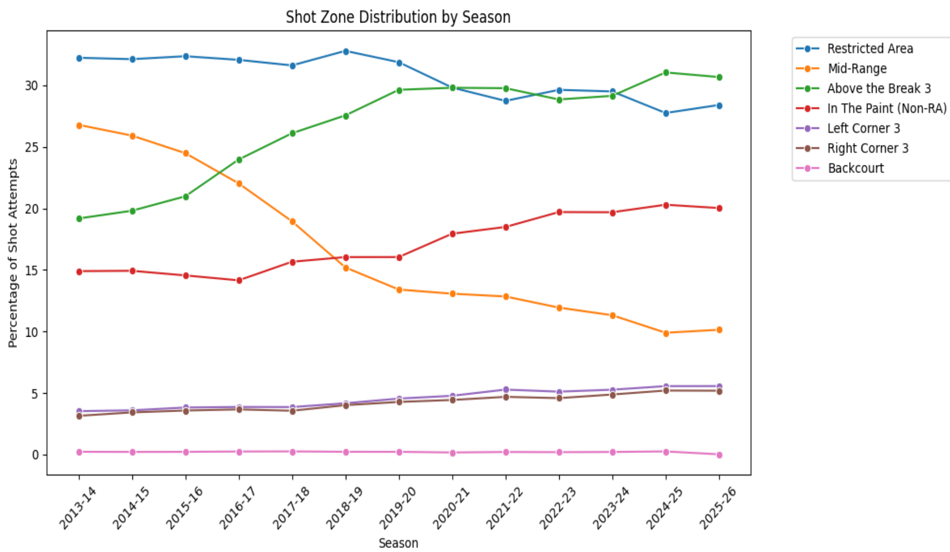
Appendix A: *Average shot distance by season, 2013-14 through 2025-26.*



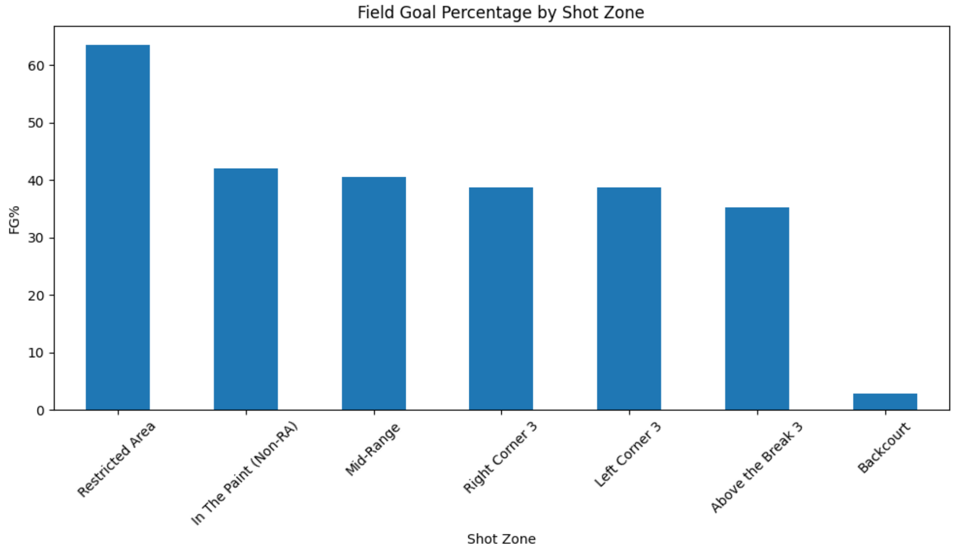
Appendix B: *Share of attempts by shot type (2PT vs 3PT) over time.*



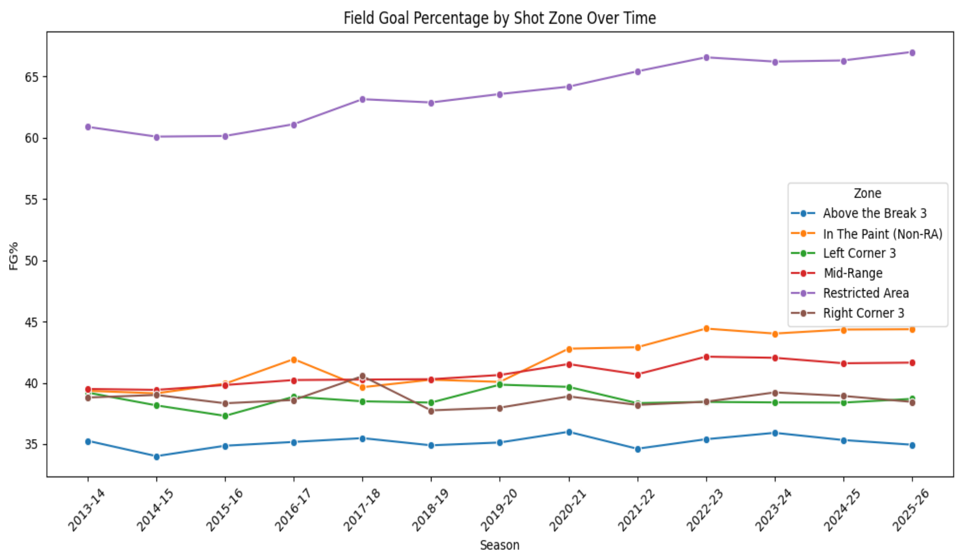
Appendix C: *Field goal percentage by shot type (2PT vs 3PT) over time.*



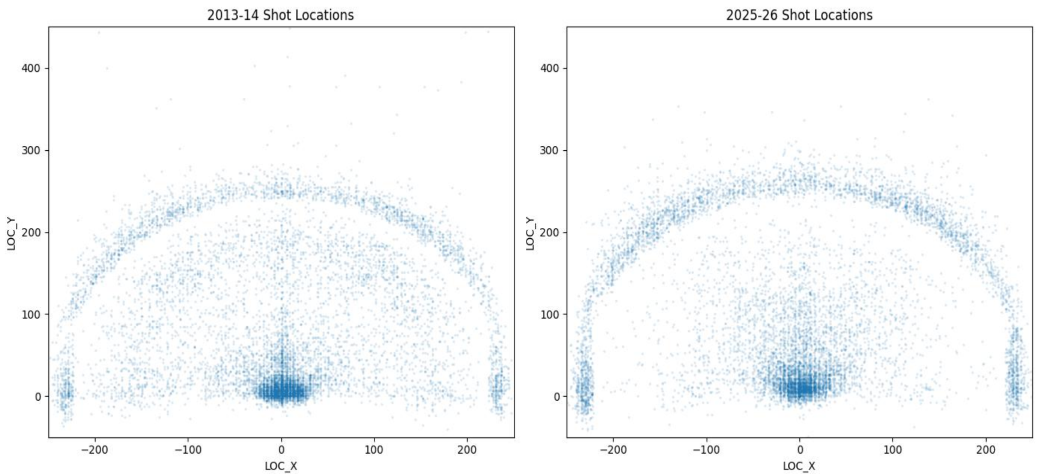
Appendix D: *Share of attempts by shot zone over time.*



Appendix E: *Field goal percentage by shot zone (aggregate 2013-present).*



Appendix F: *Field goal percentage by shot zone over time.*



Appendix G: *Shot locations 2013-14 vs 2025-26, side-by-side scatter of 10,000 sampled shots from each season.*